



Unsupervised Classification of Winter Atmospheric Regimes at Dumont d'Urville, East Antarctica

A k -means Approach Based on Radiosonde Profiles (2020–2025)

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Abstract The coastal Antarctic station Dumont d'Urville (DDU) experiences a rich variety of wintertime atmospheric regimes driven by katabatic drainage, synoptic intrusions, and radiative decoupling. This study presents an unsupervised classification of 579 austral winter (June–August) radiosondes launched at DDU between 2020 and 2025 at 00:00 UTC. Vertical profiles of temperature, dew point, dew-point depression, and horizontal wind components are concatenated and normalised using a hybrid scheme: global min-max scaling for thermodynamic variables and block z-score scaling for wind components, encoding the physical hypothesis that wind structure is the primary organiser of DDU winter regimes while preserving thermodynamic shape information. The first four principal components explain 90.8% of the variance. The silhouette score is maximised at $k = 2$ ($s = 0.30$), which separates active from passive wind regimes but provides insufficient physical resolution for documenting DDU winter diversity. Silhouette peak analysis over 500 random seeds identifies $k = 6$ as the most frequent local secondary optimum (176/500 seeds), ahead of $k = 8$ (163/500) and $k = 5$ (158/500); $k = 6$ is therefore retained as the primary result. Bootstrap co-occurrence analysis confirms that 96.2% of unique sounding pairs are stably assigned. A wind-only partition yields an Adjusted Rand Index of 0.718, indicating that the six regimes are predominantly, though not exclusively, kinematically defined. The regimes are: humid katabatic–synoptic coupling (C0), weak-wind dry state (C1), katabatic–synoptic coupling (C2), decoupled layer (C3), humid moderate-wind regime (C4), and dry meridional outflow (C5). A two-stage thermodynamic refinement is presented in the appendix and yields eleven sub-clusters.

Contents

1	Introduction	4
2	Data and Preprocessing	4
2.1	Site, instrumentation, and temporal selection	4
2.2	Quality control	4
2.3	Vertical interpolation and coverage filtering	5
3	Feature Construction and Normalisation	5
3.1	Input variables and feature matrix	5
3.2	Normalisation strategy	6
3.3	Dimensionality reduction via PCA	6
4	Physical Diagnostics	7
4.1	Thermal structure	7
4.2	Wind components and speed	7
4.3	Katabatic jet diagnostics	8
4.4	Humidity diagnostics	8
5	Clustering: Selection of k	9
5.1	Algorithm	9
5.2	The silhouette curve: $k = 2$ as the global optimum	9
5.3	Silhouette peak analysis across 500 seeds	10
5.4	Seed selection and final silhouette	11
6	Bootstrap Stability	11
7	The Six Atmospheric Regimes	12
7.1	PCA loadings: wind dominance as expected from normalisation	12
7.2	Cluster descriptions	12
8	Wind-Only Partition and ARI	18
9	Temporal and Inter-Annual Analysis	19
10	Discussion	20
10.1	Physical coherence of the six-regime taxonomy	20
10.2	On the choice of k	22
10.3	Dependence on normalisation and interpretive limitations	22
11	Limitations and Perspectives	23
11.1	Limitations	23
11.2	Perspectives	24
12	Conclusion	24
13	Appendix	27

13.1 Two-Stage Thermodynamic Refinement	27
13.1.1 Procedure	27
13.1.2 Results	27

1 Introduction

Dumont d’Urville (DDU; 66.66°S, 140.00°E) is the French permanent Antarctic station on the Petrel Island, at the foot of the East Antarctic ice sheet escarpment. Terre Adélie records some of the strongest sustained surface winds on Earth, with winds exceeding 315km h^{-1} . These extreme conditions arise from three competing physical forcings. First, gravitational drainage of cold, dense continental air from the plateau feeds a quasi-permanent katabatic jet constrained by local orography to the $90^\circ\text{--}180^\circ$ surface wind sector [3]. Second, extratropical cyclones from the Southern Ocean storm track approach from the west and inject maritime air masses into the coastal boundary layer [4]. Third, under quiescent conditions, the surface layer decouples radiatively from the free troposphere, building strong near-surface inversions during the polar night.

These regimes matter for several reasons. Katabatic winds drive coastal polynya formation with consequences for sea-ice and deep-water production [5]. Low-level sublimation within the dry katabatic layer removes up to 35% of coastal snowfall [6]. Long-term changes in synoptic intrusion frequency affect moisture transport and the surface mass balance of the ice sheet.

Routine radiosonde launches at DDU sample this variability at high vertical resolution. Unsupervised clustering offers an objective classification that lets dominant regimes emerge from the data without imposing a priori thresholds. Blount et al. [2] demonstrated the potential of *k*-means on observed thermodynamic profiles for boundary-layer regime classification; the present study extends this approach by incorporating wind profiles, which are the primary physical discriminant at DDU, alongside the thermodynamic variables. This study applies *k*-means clustering to six winters of 00:00 UTC soundings (2020–2025; 10:00 local time at DDU), with a bootstrap stability validation and a temporal analysis of the clusters.

The report follows the chronological order of the analysis: data and preprocessing (Section 2), feature construction and normalisation (Section 3), clustering and selection of *k* (Section 5), bootstrap stability (Section 6), the six regimes (Section 7), wind-only validation (Section 8), temporal analysis (Section 9), two-stage refinement (Section 13.1), and discussion (Section 10).

2 Data and Preprocessing

2.1 Site, instrumentation, and temporal selection

DDU is operated year-round by IPEV and Météo-France. Radiosonde data are publicly available through the Météo-France open data portal [1]. The analysis is restricted to June–August (JJA) and to 00:00 UTC launches only. This choice is deliberate: 00:00 UTC corresponds to 10:00 local time at DDU (UTC+10), a phase of the diurnal katabatic cycle that is relatively stationary.

2.2 Quality control

Fragment stitching is applied first: consecutive records sharing the same launch timestamp (year, month, day, hour) and whose altitude ranges are continuous within a tolerance of 500 m are merged into a single profile. This step precedes all quality control so that spike detection and

pressure monotonicity checks act on the full merged ascent.

Physical threshold checks then flag out-of-range values (temperature and dew point $\in [150, 350]$ K; pressure $\in [500, 110\,000]$ Pa; wind speed $\in [0, 200]$ m s⁻¹; wind direction $\in [0^\circ, 360^\circ]$) and enforce the physical constraint $T_d \leq T$. Duplicate levels (identical altitude) are removed. Flagged values are set to NaN rather than triggering sounding-level rejection.

Temperature spikes are detected on the valid sub-profile using a central finite-difference gradient. Levels separated by fewer than 50 m are excluded from gradient computation to avoid artefacts from near-coincident levels. A level is flagged if its gradient exceeds $+0.1$ K m⁻¹ (superadiabatic inversion, physically implausible) or falls below $-3\Gamma_d \approx -29$ K km⁻¹ (three times the dry adiabatic lapse rate). Wind spikes are detected analogously: a level is flagged if the wind-speed shear exceeds 0.15 m s⁻¹ m⁻¹ (150 m s⁻¹ km⁻¹), with the same 50 m minimum-separation guard. In both cases only the upper level of the offending pair is set to NaN, preserving the surrounding valid data.

Pressure monotonicity is enforced by a forward-greedy scan: the first level is retained, and each subsequent level is kept only if its pressure is strictly lower than the last retained level. Non-monotonic levels (pressure inversions in the column) are removed rather than flagged. A sounding is discarded entirely if its pressure column contains a consecutive gap of more than five missing levels, or if fewer than 20 valid levels remain after the above steps; 20 soundings were discarded on these criteria.

2.3 Vertical interpolation and coverage filtering

Each retained sounding is linearly interpolated onto the uniform altitude grid

$$z_{\text{grid}} = \{50, 75, \dots, 5975\}\text{m} \quad (1)$$

($L = 238$ levels, step 25 m). Wind components are interpolated in Cartesian form (u , v) and speed and direction recomputed afterwards. A sounding is then discarded if its lowest valid altitude exceeds 50 m or if it does not reach 85% of the grid top (≈ 4300 m); 167 soundings were discarded on this criterion. Combined with the 21 discarded during quality control, a total of 188 soundings were removed (21.8%), leaving **579** profiles. Residual missing values after interpolation are imputed by the per-level mean.

3 Feature Construction and Normalisation

3.1 Input variables and feature matrix

Five vertical variables are extracted: temperature T , dew-point T_d , dew-point depression $T - T_d$, and wind components $u = -ff \sin(\text{dd})$, $v = -ff \cos(\text{dd})$. The feature matrix is

$$\mathbf{X} \in \mathbb{R}^{579 \times 5L}, \quad \mathbf{X}_i = (T_i(z_1), \dots, T_i(z_L), T_{d,i}(z_1), \dots, (T - T_d)_i(z_1), \dots, u_i(z_1), \dots, v_i(z_L)). \quad (2)$$

3.2 Normalisation strategy

The goal is to cluster the *vertical shape* of profiles rather than their absolute amplitude, since surface inversion structure, jet height, and humidity gradient are the physically relevant discriminants. A two-step hybrid normalisation is applied, treating the thermodynamic and kinematic variables differently, for physically motivated reasons.

Step 1 — Global min-max scaling of thermodynamic variables. Temperature T , dew-point T_d , and dew-point depression $T - T_d$ are mapped to $[0, 1]$ by global min-max scaling:

$$\tilde{X}_{i,j}^{(\text{therm})} = \frac{X_{i,j} - \min_j}{\max_j - \min_j}, \quad (3)$$

where $\min_j$ and $\max_j$ are computed over all soundings and all levels of variable j . This global scaling preserves the relative shape of vertical profiles (inversion strength, humidity gradients) while removing the influence of the mean temperature level, which varies with season and synoptic conditions but is not the primary discriminant of tropospheric layer regime at DDU.

Step 2 — Block z-score scaling of wind components. The wind components u and v are standardised to zero mean and unit variance within each variable block:

$$\tilde{X}_{i,j}^{(\text{wind})} = \frac{X_{i,j} - \mu_j}{\sigma_j}, \quad (4)$$

where μ_j and σ_j are the mean and standard deviation computed over all soundings and levels of variable j .

Physical justification. The two-step design encodes a physical hypothesis: wind direction and vertical wind structure are assumed to be the *primary* discriminants of atmospheric regimes at DDU in winter, while thermodynamic variables carry complementary shape information. The block z-score scaling of u and v preserves the directional and vertical wind structure while giving the wind variables a larger effective variance than the min-max scaled thermodynamics. After normalisation, the block statistics confirm this design: u has standard deviation $\sigma = 1.22$ and v has $\sigma = 1.03$, while T , T_d , and $T - T_d$ have standard deviations of 0.12, 0.13, and 0.17 respectively. Wind components therefore contribute more variance to the PCA, which is intentional: it reflects the assumption that the large-scale wind structure is the leading organiser of DDU winter low tropospheric layer regimes.

3.3 Dimensionality reduction via PCA

PCA is applied to $\tilde{\mathbf{X}}$. table 1 reports the variance decomposition. The first four components explain 90.8% of the total variance (PC1: 47.3%; PC2: 36.1%; PC3: 5.0%; PC4: 2.5%), satisfying the $\geq 90\%$ threshold adopted here. All clustering is performed on $\mathbf{X}_{\text{red}} \in \mathbb{R}^{579 \times 4}$.

Table 1: Explained variance of the first ten principal components ($n = 579$, $5L = 1190$ features after normalisation).

Component	Individual (%)	Cumulative (%)
PC1	47.3	47.3
PC2	36.1	83.4
PC3	5.0	88.4
PC4	2.5	90.8
PC5	1.8	92.6
PC6	1.2	93.8
PC7	0.8	94.5
PC8	0.6	95.2
PC9	0.6	95.8
PC10	0.5	96.3

4 Physical Diagnostics

The clustering procedure operates on normalised concatenated reduced profiles and produces abstract cluster labels. To attach physical meaning to these labels, 40 scalar diagnostics are computed from the original (non-normalised) profiles of each sounding and aggregated per cluster (mean and standard deviation). These diagnostics are independent of the clustering procedure; they are used purely a posteriori for interpretation. This section defines the principal diagnostics explicitly, grouped by physical category. All altitudes are expressed relative to the lowest valid level of each sounding ($z_{\text{rel}} = z - z_0$).

4.1 Thermal structure

Lapse rates. Layer lapse rates are estimated by ordinary least-squares regression of T on z within each layer:

$$\Gamma_{[z_1, z_2]} = \hat{\beta}_1, \quad T(z) \approx \hat{\beta}_0 + \hat{\beta}_1 z, \quad z \in [z_1, z_2], \quad (5)$$

computed with a minimum of three valid levels. Positive values indicate a temperature inversion; negative values a normal lapse. The dry adiabatic lapse rate is $\Gamma_d = 9.8 \text{ K km}^{-1}$. A fine profile of lapse rates is also computed in contiguous 200 m windows from 0 to 2000 m (i.e. $[0, 200]$, $[200, 400]$, $\dots$, $[1800, 2000]$ m),

4.2 Wind components and speed

Surface and layer winds. The surface wind speed $w_{s_{\text{sfc}}}$ is the wind speed at the lowest valid level of the profile. Surface wind components ($u_{s_{\text{sfc}}}$, $v_{s_{\text{sfc}}}$) are the layer mean over the lowest 200 m (minimum three valid levels required). Wind speed and components at 500 m, 1 km, and 3 km are obtained by linear interpolation of the respective profiles onto those target altitudes. Layer-mean wind speeds over $[0, 1 \text{ km}]$ and $[1, 3 \text{ km}]$ are simple vertical averages (minimum three valid levels).

Wind turning. The wind turning angle between the surface and 3 km measures the rotation of the wind vector with altitude, and is indicative of thermal wind and synoptic advection:

$$\Delta\alpha_{0\rightarrow 3\text{km}} = \arg\{\exp[i(\arctan 2(-u_{3\text{km}}, -v_{3\text{km}}) - \arctan 2(-u_{\text{sfc}}, -v_{\text{sfc}}))]\} \times \frac{180}{\pi}, \quad (6)$$

expressed in degrees. The complex exponential formulation ensures the result lies in $(-180^\circ, +180^\circ]$. Positive (clockwise) turning indicates warm-air advection (veering); negative (anticlockwise) turning indicates cold-air advection (backing).

4.3 Katabatic jet diagnostics

Jet nose speed and height. The katabatic jet is identified as the wind speed maximum in the lowest kilometre:

$$ws_{\text{jet}} = \max_{z_{\text{rel}} \leq 1\text{km}} ws(z), \quad z_{\text{jet}} = \arg \max_{z_{\text{rel}} \leq 1\text{km}} ws(z). \quad (7)$$

A low jet nose ($z_{\text{jet}} < 300\text{m}$) is characteristic of intense katabatic drainage under strong stable stratification. A higher nose indicates a more diffuse or synoptically modulated flow.

Jet shear. The shear below the jet nose (from surface to nose) and above (from nose to 1 km) quantify the vertical wind structure:

$$\delta ws_{\text{below}} = ws_{\text{jet}} - ws_{\text{sfc}}, \quad \delta ws_{\text{above}} = ws_{1\text{km}} - ws_{\text{jet}}. \quad (8)$$

Negative δws_{above} indicates a well-defined jet with clear deceleration above the nose.

4.4 Humidity diagnostics

Dew-point depression. The dew-point depression is the difference between temperature and dew-point, a direct measure of the unsaturation of the air:

$$\delta T_d(z) = T(z) - T_d(z). \quad (9)$$

Dew-point values below the sensor floor of approximately -60°C (213.15K) are masked as unreliable. Layer means are computed over $[0, 500\text{m}]$ and $[1, 3\text{km}]$.

Relative humidity. Relative humidity is computed using the Magnus–Tetens approximation for the saturation vapour pressure:

$$e_s(T) = 6.112 \exp\left(\frac{17.67(T - 273.15)}{T - 273.15 + 243.5}\right) \text{hPa}, \quad (10)$$

giving

$$RH(z) = 100 \frac{e_s(T_d(z))}{e_s(T(z))} \%. \quad (11)$$

Layer means $RH_{[0,1\text{km}]}$ and $RH_{[1,3\text{km}]}$, the column maximum RH_{max} , and its altitude $z_{RH_{\text{max}}}$ are reported.

Integrated water vapour. The integrated water vapour (IWV, also called precipitable water) is obtained by vertical integration of the specific humidity q in pressure coordinates:

$$q(z) = \frac{0.622 e_s(T_d(z))}{p(z)/100 - 0.378 e_s(T_d(z))}, \quad (12)$$

$$\text{IWV} = \left| \int_{\text{top}}^{\text{sfc}} q \frac{dp}{g} \right| \approx \left| \sum_i q_i \frac{\Delta p_i}{g} \right| \text{ kg m}^{-2}, \quad (13)$$

where $g = 9.81 \text{ m s}^{-2}$ and the trapezoidal rule is used for numerical integration over available pressure levels. IWV is the most integrative humidity metric and the primary discriminant between humid (maritime intrusion, $\text{IWV} \gtrsim 4 \text{ kg m}^{-2}$) and dry (continental drainage, $\text{IWV} \lesssim 2 \text{ kg m}^{-2}$) regimes at DDU.

5 Clustering: Selection of k

5.1 Algorithm

K -means minimises the within-cluster inertia

$$\mathcal{I} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2. \quad (14)$$

The number of clusters is guided by the mean silhouette score

$$\bar{s} = \frac{1}{n} \sum_{i=1}^n \frac{b_i - a_i}{\max(a_i, b_i)}, \quad (15)$$

where a_i is the mean intra-cluster distance and b_i the mean distance to the nearest other cluster.

5.2 The silhouette curve: $k = 2$ as the global optimum

Figure 1 shows the silhouette curve for one representative seed. The global maximum occurs at $k = 2$ ($\bar{s} = 0.30$). This result is robust: over 500 seeds, $k = 2$ is the highest silhouette value in 415/500 runs (83%). It is also physically interpretable: $k = 2$ separates *active* from *passive* wind regimes, grouping soundings with strong surface or upper-level flow against those with weak winds at all levels. This binary partition is real and meaningful, but it provides insufficient physical resolution to document the diversity of DDU winter atmospheric states. A single active-wind cluster conflates the intense katabatic-synoptic coupling, the moderate katabatic, and the synoptic zonal intrusion, three physically and meteorologically distinct states with fundamentally different moisture contents and tropospheric structures.

The silhouette score decreases monotonically after $k = 2$: this is a structural property of the silhouette metric, which favours large and compact clusters. Using the global maximum alone would therefore stop the analysis at a coarser level than the data support. We instead seek *local maxima* of the silhouette curve, which indicate values of k at which genuine additional structure appears above the decreasing dry trend.

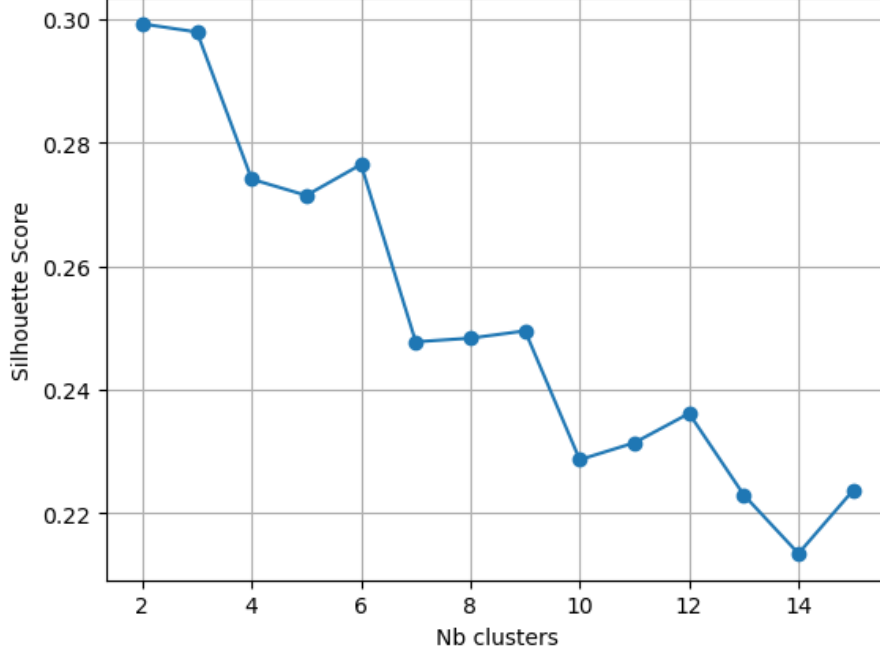


Figure 1: Silhouette score as a function of k (seed 24, $n_{\text{init}} = \text{auto}$, $k \in [2, 15]$). The global maximum at $k = 2$ ($\bar{s} = 0.30$) reflects the coarse active/passive wind dichotomy. The silhouette decreases monotonically after $k = 2$; local peaks at $k = 6$, $k = 8$, and $k = 5$ indicate the presence of finer genuine structure.

5.3 Silhouette peak analysis across 500 seeds

Since the silhouette curve depends on the random seed, we compute it over 500 independent seeds and count, for each k , how often it constitutes a local maximum (i.e. $\bar{s}(k) > \bar{s}(k - 1)$ and $\bar{s}(k) > \bar{s}(k + 1)$). table 2 reports the results. The most frequent local peak is $k = 6$ (176/500 seeds, 35%), followed by $k = 8$ (163/500, 33%) and $k = 5$ (158/500, 32%). These three values are statistically close; no single secondary optimum dominates clearly. Crucially, $k = 4$ is the second-most frequent *global* best (85/500 seeds after $k = 2$) but appears as a local peak in only 123/500 seeds, meaning it often benefits from the decreasing trend rather than indicating genuinely new structure. We retain $k = 6$ as the primary result, as it is the most frequent local peak and therefore the value at which genuine additional cluster structure most consistently appears.

Table 2: Silhouette peak analysis over 500 seeds ($k \in [2, 15]$, $n_{\text{init}} = \text{auto}$). A “local peak” means $\bar{s}(k) > \bar{s}(k \pm 1)$. “Global best” is the number of seeds for which k achieves the highest silhouette value.

k	Local peak	Global best	Note
2	—	415/500	Global maximum; not a local peak by definition
6	176/500	—	Most frequent local peak; retained
8	163/500	—	Second local peak
5	158/500	—	Third local peak
4	123/500	85/500	Partly rides the decreasing trend from $k = 2$

5.4 Seed selection and final silhouette

The random seed is fixed by exhaustive search over seeds 0–199 with $n_{\text{init}} = \text{auto}$ (k-means++ initialisation): the seed maximising the silhouette for $k = 6$ is retained. This yields **seed = 24** with $\bar{s} = \mathbf{0.2765}$. The final labels are assigned by `fit_kmeans(k=6, seed=24, n_init='auto')`.

6 Bootstrap Stability

To quantify dependence on random initialisation, k -means is run $R = 200$ times (seeds 0–199, $n_{\text{init}} = 20$). The co-occurrence matrix is

$$C_{ij} = \frac{1}{R} \sum_{r=1}^R \mathbf{1}[\ell_i^{(r)} = \ell_j^{(r)}], \quad (16)$$

where $\ell_i^{(r)}$ is the label of sounding i in run r . Diagonal entries ($C_{ii} = 1$) and double-counting are removed; table 3 reports corrected counts over the 167 331 unique sounding pairs ($579 \times 578/2$).

Table 3: Bootstrap co-occurrence statistics ($R = 200$, seeds 0–199, $n_{\text{init}} = 20$). Counts are corrected for the symmetric matrix and diagonal. Total unique pairs: 167 331.

Category	Unique pairs	%
Co-assigned ($C_{ij} > 0.8$)	30 226	18.1%
Separated ($C_{ij} < 0.2$)	130 751	78.1%
Stable or separated (subtotal)	160 977	96.2%
Ambiguous ($0.3 \leq C_{ij} \leq 0.7$)	3 992	2.4%

The distribution is strongly bimodal: 96.2% of pairs are either stably co-assigned or stably separated. Only 2.4% fall in the ambiguous range, corresponding to approximately 65 boundary soundings located near cluster frontiers in PCA space.

7 The Six Atmospheric Regimes

7.1 PCA loadings: wind dominance as expected from normalisation

The PCA loadings are the components of the eigenvectors of the covariance matrix of $\tilde{\mathbf{X}}$: the loading of feature j on principal component p is the coefficient w_{jp} such that

$$\text{PC}_p = \sum_{j=1}^{5L} w_{jp} \tilde{X}_j, \quad (17)$$

where $\tilde{X}_j$ is the j -th column of the normalised feature matrix.

The loading structure (Figure 2) is consistent with what the normalisation design predicts. Since u and v have post-normalisation standard deviations of 1.22 and 1.03 respectively, compared to 0.12–0.17 for the thermodynamic variables, wind components carry more variance into the PCA and naturally dominate the leading components. This is a validation that the normalisation is working as intended. Specifically: PC1 is dominated by v at upper levels (above 5750 m, loadings ≈ 0.073), reflecting the meridional large-scale flow; PC2 by u at upper levels (above 5650 m, loadings ≈ 0.068), reflecting the zonal flow; PC3 by u around 1100–1300 m (loadings ≈ 0.099), which corresponds to the typical katabatic jet nose height. PC4, the fourth component needed to reach 90% of explained variance, returns to v at upper levels, suggesting a second meridional mode. The thermodynamic variables do not appear among the dominant loadings of any of the four retained components, confirming that they act as secondary descriptors within the wind-defined regime space.

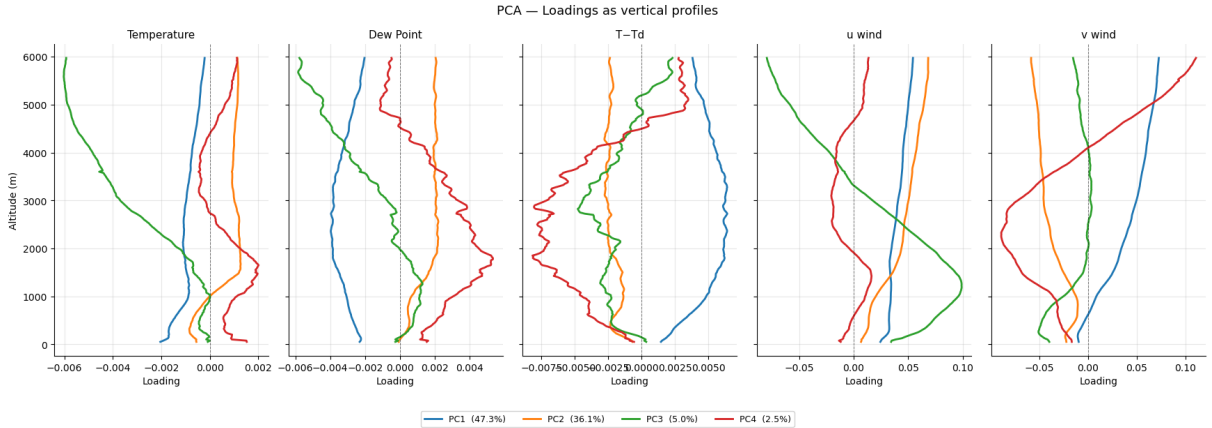


Figure 2: PCA loadings as vertical profiles for PC1–PC3, one panel per input variable. PC1 and PC2 are dominated by u and v at all levels. Thermodynamic variables contribute mainly to PC3.

7.2 Cluster descriptions

Figure 3 shows the scatterplot of PCA1 and PCA2 colored by cluster. The cluster forms a continuum as the profiles are the representation of low troposphere regimes.

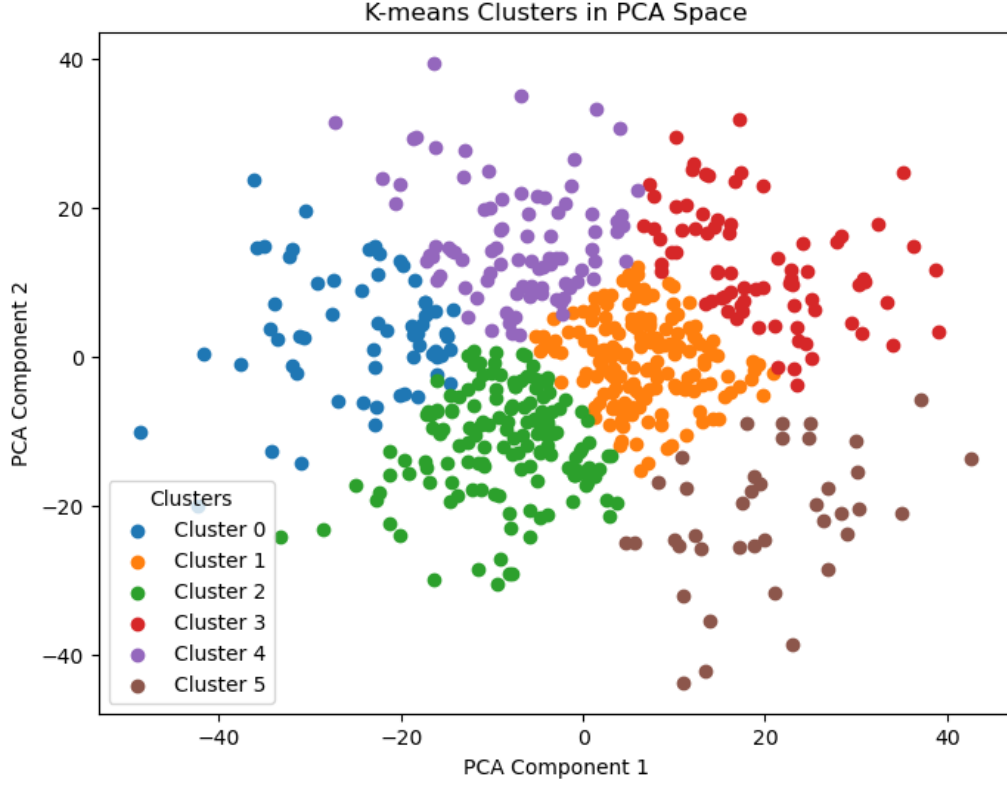


Figure 3: Scatterplot of PCA1 vs PCA2 colored by cluster, seed=24, k=6

The table 4 summarises the six clusters (seed 24) with significant features and mean vertical profiles are shown in Figure 4.

Table 4: Physical diagnostics of the six k -means clusters ($k = 6$, seed 24, $n = 579$). ws_{sfc} : surface wind speed; ws_3 : wind speed at 3 km; $\overline{RH}_{0-1}$: mean relative humidity 0–1 km; IWV: integrated water vapour; u_3/v_3 : wind components at 3 km.

Cluster	Label	ws_{sfc} (m s^{-1})	ws_3 (m s^{-1})	$\overline{RH}_{0-1}$ (%)	IWV (kg m^{-2})	u_3 (m s^{-1})	v_3 (m s^{-1})	n
C0	HKS	15.3	13.9	72	5.05	−3.6	−12.0	63
C1	WWD	7.2	10.0	42	1.74	+8.2	+3.3	169
C2	KSC	14.7	8.0	51	2.68	−5.2	+1.4	144
C3	DL	5.2	20.3	41	1.97	+19.1	+4.2	77
C4	HMW	8.1	14.5	67	4.06	+10.1	−9.1	87
C5	DMO	9.6	18.3	39	1.03	+2.9	+16.8	39

Labels: HKS = Humid Katabatic–Synoptic coupling; WWD = Weak-Wind Dry state; KSC = Katabatic–Synoptic Coupling; DL = Decoupled Layer; HMW = Humid Moderate Wind; DMO = Dry Meridional Outflow.

Statistical characterisation of cluster separation

Table 5 reports the standardised effect size $z_{k,v} = (\bar{x}_{k,v} - \bar{x}_v)/\sigma_v$ for each cluster k and variable v , where $\bar{x}_v$ and σ_v are the mean and standard deviation computed over all 579 soundings. A value $|z| > 0.5$ indicates that the cluster mean departs by more than half a global standard deviation from the dataset average, and is used here as the threshold for a physically meaningful separation. Table 6 reports the one-way ANOVA F -statistic and p -value for each variable, quantifying whether the between-cluster variance is significant relative to the within-cluster variance.

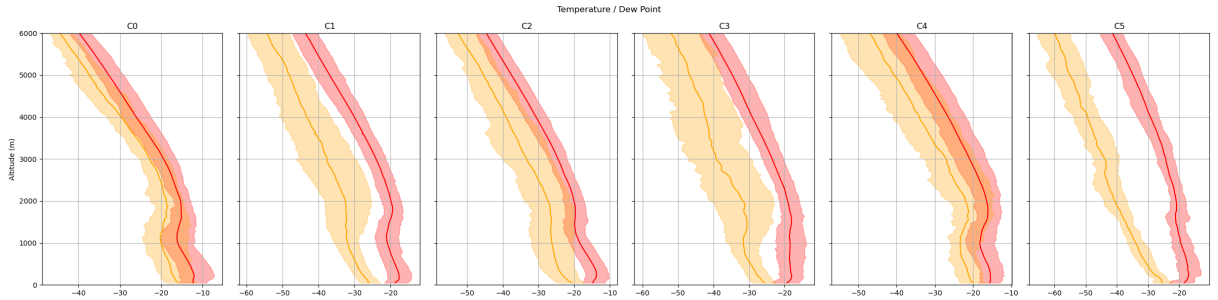
Table 5: Standardised effect size $z_{k,v} = (\bar{x}_{k,v} - \bar{x}_v)/\sigma_v$ for the six clusters and the most discriminating variables. Bold: $|z| > 1.0$. Shading: positive (warm) / negative (cool). Variables with no $|z| > 0.5$ across all clusters are omitted.

Variable	C0	C1	C2	C3	C4	C5
<i>Wind components</i>						
u_{sfc}	-1.22	0.39	-0.61	0.99	-0.01	0.58
v_{sfc}	0.25	-0.32	0.71	-0.42	-0.48	0.25
u_1	-1.21	0.28	-0.50	1.21	0.13	-0.08
u_3	-0.86	0.33	-1.02	1.41	0.52	-0.21
v_3	-1.40	0.34	0.12	0.44	-1.06	1.87
ws_{sfc}	0.78	-0.41	0.69	-0.70	-0.27	-0.05
ws_{500}	1.24	-0.46	0.49	-0.78	-0.14	0.02
ws_3	0.20	-0.38	-0.69	1.17	0.29	0.87
wind_turning	-0.95	0.59	-0.34	0.66	-0.63	0.33
jet_nose_speed	1.04	-0.45	0.57	-0.76	-0.28	0.31
<i>Thermodynamic</i>						
T_{sfc}	0.72	-0.35	0.35	-0.43	0.09	-0.27
θ_{sfc}	0.79	-0.38	0.41	-0.49	0.08	-0.36
N_{sfc}^2	-0.39	0.03	-0.08	0.50	-0.27	0.44
$\Gamma_{0-2\text{ km}}$	-0.05	0.23	-0.77	0.57	0.50	-0.30
$\Gamma_{400-600}$	-0.16	0.03	-0.44	0.76	-0.08	0.41
$\Gamma_{600-800}$	-0.28	0.08	-0.51	0.97	0.04	-0.04
$\Gamma_{800-1000}$	-0.18	0.02	-0.53	0.75	0.39	-0.19
$\Gamma_{1800-2000}$	0.18	-0.09	-0.02	-0.04	0.01	0.23
<i>Humidity</i>						
IWV	1.31	-0.51	0.01	-0.38	0.76	-0.90
$\overline{RH}_{0-1}$	1.06	-0.42	-0.01	-0.51	0.79	-0.58
$\overline{RH}_{1-3}$	0.95	-0.46	0.31	-0.48	0.62	-1.13
δT_d^{1-3}	-0.80	0.37	-0.29	0.45	-0.53	1.10

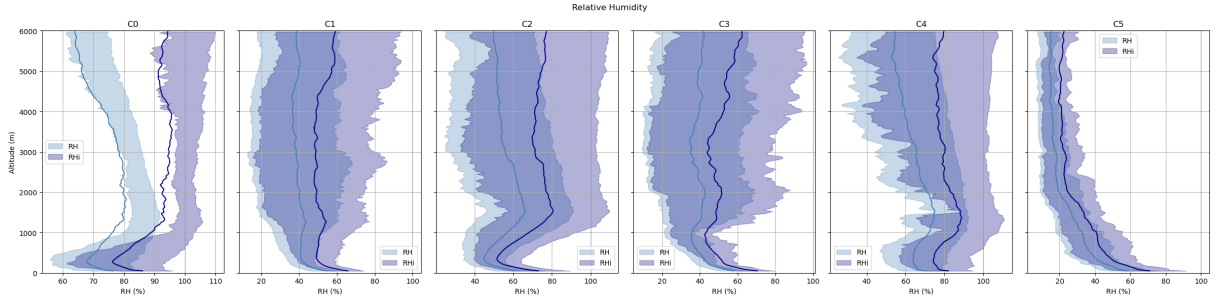
Table 6: One-way ANOVA results for each diagnostic variable across the six clusters ($n = 579$). A large F indicates strong between-cluster separation relative to within-cluster variance. Significance: *** $p < 0.001$, * $p < 0.05$.

Variable	F	p	
v_3	244.3	< 0.001	***
u_3	244.0	< 0.001	***
u_{sfc}	95.4	< 0.001	***
IWV	84.3	< 0.001	***
ws_3	79.2	< 0.001	***
$\overline{RH}_{1-3}$	64.2	< 0.001	***
wind_turning	62.7	< 0.001	***
jet_nose_speed	62.7	< 0.001	***
$\overline{RH}_{0-1}$	55.1	< 0.001	***
ws_{sfc}	51.6	< 0.001	***
v_{sfc}	32.7	< 0.001	***
$\Gamma_{600-800}$	28.8	< 0.001	***
$\Gamma_{800-1000}$	24.2	< 0.001	***
T_{sfc}	21.0	< 0.001	***
$\Gamma_{400-600}$	18.5	< 0.001	***
N_{sfc}^2	9.4	< 0.001	***
jet_nose_height	9.2	< 0.001	***
$\Gamma_{0-2 \text{ km}}$	38.0	< 0.001	***
u_1	90.8	< 0.001	***
θ_{sfc}	27.4	< 0.001	***
δT_d^{1-3}	45.1	< 0.001	***
Γ_{0-200}	5.0	< 0.001	***
$\Gamma_{1800-2000}$	1.2	0.331	—

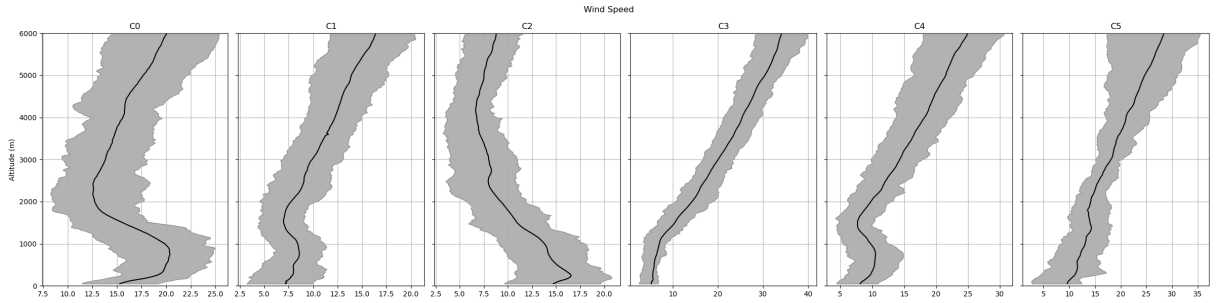
Several observations follow directly from these tables. First, $\Gamma_{1800-2000}$ is the only variable that does not discriminate the clusters ($F = 1.2$, $p = 0.33$), confirming that the free-tropospheric thermal structure above 1800 m is identical across all regimes and validating the choice to restrict the analysis to the lowest 6 km. All other variables separate the clusters at $p \leq 0.001$, confirming that the partition captures statistically significant structure across a broad range of physical diagnostics, including those given lower weight in the normalisation scheme. Second, u_3 and v_3 yield the highest F -values ($F = 244$), consistent with the normalisation design that gives wind components dominant weight in PCA space; u_1 ($F = 90.8$) is nearly as discriminating, highlighting the 1 km level as a key kinematic discriminant. Third, IWV ($F = 84.3$) and δT_d^{1-3} ($F = 45.1$) achieve high F -values despite receiving lower weight in the clustering via the min-max normalisation, confirming that the kinematic partition also separates thermodynamically distinct air masses. Fourth, the lapse-rate variables in the 400–1000 m layer ($F = 18$ –29) and the integrated lapse rate 0–2 km ($F = 38$) are significantly discriminating despite their large within-cluster standard deviations, indicating that between-cluster differences in thermal structure are real even if individual soundings vary widely within each regime. C1 is the only cluster with no $|z| > 0.5$ on any variable (Table 5), confirming quantitatively that it sits close to the dataset mean on all diagnostics and is defined by the absence of a strong forcing signature rather than by a positive physical signal. This observation is validated by the position of C1 points in the PCA space in Figure 3.



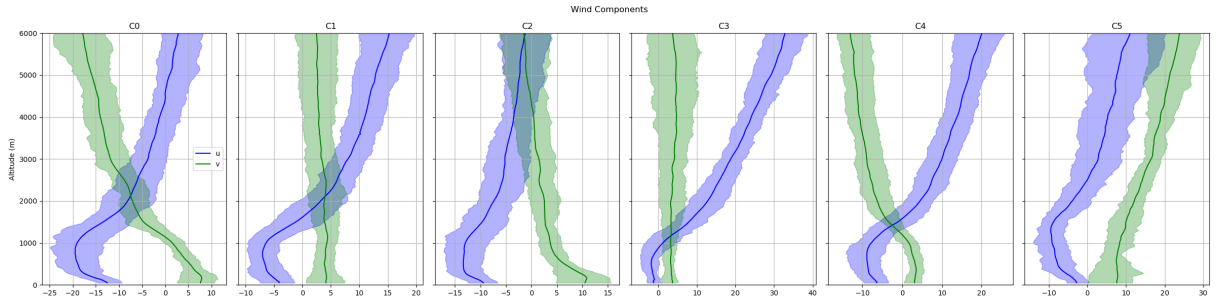
(a) Temperature (T , red) and dew-point temperature (T_d , orange) profiles for the six clusters.



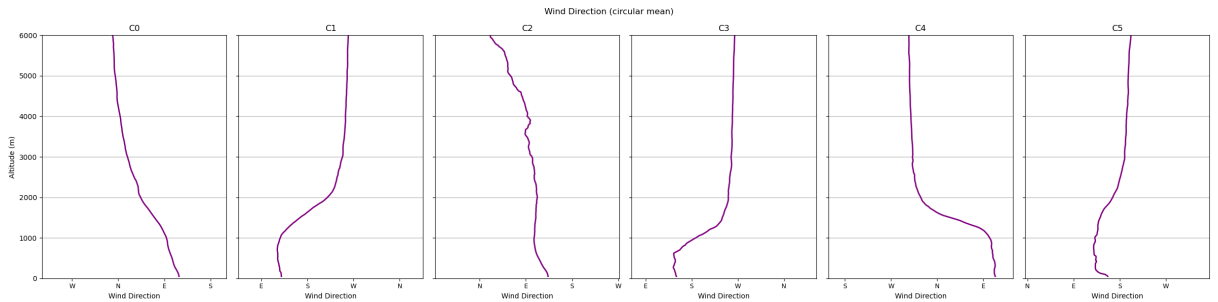
(b) Relative humidity (RH and RH_i, dark and light blue respectively) profiles for the six clusters.



(c) Wind-speed (WS) profiles for the six clusters.



(d) Zonal (u , blue) and meridional (v , green) wind-component profiles for the six clusters.



(e) Wind-direction (dd) profiles for the six clusters.

Figure 4: Cluster-mean vertical profiles (50–6000 m) for the six clusters. Panels show temperature and dew point (T , T_d), relative humidity (RH), wind speed (WS), zonal and meridional wind components (u , v), and wind direction (dd). Shaded ribbons indicate the 25th–75th percentile range.

C0 — Humid katabatic–synoptic coupling ($n = 63$, $\text{IWV} = 5.05 \text{ kg m}^{-2}$) The most humid cluster ($\text{IWV} = 5.05 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 72\%$, dew-point depression 3.5 K in the 1–3 km layer). Strong surface wind ($\text{ws}_{\text{sfc}} = 15.3 \text{ m s}^{-1}$, $u_{\text{sfc}} = -14.7 \text{ m s}^{-1}$, $v_{\text{sfc}} = +7.7 \text{ m s}^{-1}$) with a jet nose at 592 m and a near-zero surface inversion ($\Gamma_{0-200 \text{ m}} = +0.18 \text{ K (100 m)}^{-1}$, the lowest of all clusters), consistent with mechanical mixing of the stable surface layer. The wind speed at 500 m reaches $\text{ws}_{500} = 20.1 \text{ m s}^{-1}$, indicating that the wind maximum lies above the surface and that the jet is more diffuse and elevated than in C2, this difference in vertical jet structure is the primary kinematic discriminant between the two high-wind clusters. Upper-level flow shows $u_3 = -3.6 \text{ m s}^{-1}$ and $v_3 = -12.0 \text{ m s}^{-1}$. The wind turning of $-99^\circ \pm 28^\circ$ is the most directionally consistent of all clusters, indicating a systematic backing between surface and upper levels. The surface temperature is the warmest of all clusters ($T_{\text{sfc}} = -12.2^\circ \text{C}$).

C1 — Weak-wind dry state ($n = 169$, $\text{IWV} = 1.74 \text{ kg m}^{-2}$) Weak winds at all levels ($\text{ws}_{\text{sfc}} = 7.2 \text{ m s}^{-1}$, $\text{ws}_3 = 10.0 \text{ m s}^{-1}$, $\overline{\text{ws}}_{1-3 \text{ km}} = 8.2 \text{ m s}^{-1}$) and a dry column ($\overline{\text{RH}}_{0-1} = 42\%$, $\text{IWV} = 1.74 \text{ kg m}^{-2}$). The surface inversion is moderate ($\Gamma_{0-200 \text{ m}} = +0.64 \text{ K (100 m)}^{-1}$). The wind turning of $+71^\circ \pm 97^\circ$ has a large standard deviation, indicating substantial directional heterogeneity within this cluster; any directional interpretation should be treated with caution. With $n = 169$ soundings (29% of the dataset), C1 is the largest cluster by count. Its large size combined with its internal directional heterogeneity strongly suggests that it aggregates physically distinct weakly-forced situations that the current six-cluster solution cannot resolve; it is best understood as a composite background state rather than a single coherent regime.

C2 — Katabatic–synoptic coupling ($n = 144$, $\text{IWV} = 2.68 \text{ kg m}^{-2}$) The most physically coherent cluster of the six. Strong surface flow ($\text{ws}_{\text{sfc}} = 14.7 \text{ m s}^{-1}$, $u_{\text{sfc}} = -11.0 \text{ m s}^{-1}$, $v_{\text{sfc}} = +10.6 \text{ m s}^{-1}$) with a low jet nose at 367 m and a moderately humid column ($\overline{\text{RH}}_{0-1} = 51\%$, $\text{IWV} = 2.68 \text{ kg m}^{-2}$). The thermal profile shows a surface inversion below the jet nose ($\Gamma_{0-200 \text{ m}} = +0.65 \text{ K (100 m)}^{-1}$), followed by the strongest and most statistically robust negative lapse rates of all clusters immediately above ($\Gamma_{400-600} = -0.71 \pm 0.29$, $\Gamma_{600-800} = -0.74 \pm 0.24 \text{ K (100 m)}^{-1}$, the lowest standard deviations in the entire lapse-rate table), reflecting intense and consistent thermal stratification at the jet-nose level. This is further confirmed by the strongest above-jet deceleration of all clusters ($\delta \text{ws}_{\text{above}} = -5.44 \pm 5.01 \text{ m s}^{-1}$), indicating a well-defined jet with a sharp upper boundary. The wind turning of $-31^\circ \pm 71^\circ$ indicates a moderate backing between surface and upper levels. The low spread across most variables makes this cluster the most robust and interpretable of the solution.

C3 — Decoupled layer ($n = 77$, $\text{IWV} = 1.97 \text{ kg m}^{-2}$) The calmest surface of all clusters ($\text{ws}_{\text{sfc}} = 5.2 \text{ m s}^{-1}$) with a strong upper-level flow ($\text{ws}_3 = 20.3 \text{ m s}^{-1}$, $u_3 = +19.1 \text{ m s}^{-1}$) and a dry-to-moderate column ($\text{IWV} = 1.97 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 41\%$). The surface inversion is strong ($N_{\text{sfc}}^2 = 4.5 \times 10^{-4} \text{ s}^{-2}$, the highest of all clusters), and lapse rates above 400 m are near-zero but with large standard deviations ($\Gamma_{400-600} = -0.12 \pm 0.64$, $\Gamma_{600-800} \approx 0 \pm 0.87 \text{ K (100 m)}^{-1}$), forming a quasi-isothermal yet internally heterogeneous layer between the surface and the overlying flow, in contrast to C2 where the same layer exhibits tightly constrained lapse rates. This thermodynamic heterogeneity is consistent with a decoupling that varies in depth and intensity across individual soundings. Notably, C3 shares a near-identical surface static stability with C5

($N_{\text{sfc}}^2 = 4.4 \times 10^{-4} \text{ s}^{-2}$), despite their very different upper-level flow directions, suggesting that strong surface decoupling can arise under distinct synoptic configurations. The wind turning of $+79^\circ \pm 83^\circ$ is consistent with the large difference in wind structure between surface and upper levels.

C4 — Humid moderate-wind regime ($n = 87$, $\text{IWV} = 4.06 \text{ kg m}^{-2}$) Moderately strong winds at all levels ($\text{ws}_{\text{sfc}} = 8.1 \text{ m s}^{-1}$, $\text{ws}_3 = 14.5 \text{ m s}^{-1}$) and the second most humid column ($\overline{\text{RH}}_{0-1} = 67\%$, $\text{IWV} = 4.06 \text{ kg m}^{-2}$, dew-point depression 5.6 K in the 1–3 km layer). Upper-level flow shows $u_3 = +10.1 \text{ m s}^{-1}$ and $v_3 = -9.1 \text{ m s}^{-1}$. The surface inversion is very low ($\Gamma_{0-200 \text{ m}} = +0.07 \text{ K (100 m)}^{-1}$). The wind turning of $-64^\circ \pm 129^\circ$ has a very large standard deviation, indicating that this cluster aggregates situations with substantially different directional structures; directional interpretations should be treated with caution. This cluster is less well defined than C0 and C2 and may represent a transitional regime between katabatic and synoptic forcing.

C5 — Dry meridional outflow ($n = 39$, $\text{IWV} = 1.03 \text{ kg m}^{-2}$) The driest cluster ($\text{IWV} = 1.03 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 39\%$, dew-point depression 18.5 K in the 1–3 km layer). Moderate surface wind ($\text{ws}_{\text{sfc}} = 9.6 \text{ m s}^{-1}$) with a strong upper-level southward flow ($v_3 = +16.8 \text{ m s}^{-1}$, $u_3 = +2.9 \text{ m s}^{-1}$, $\text{ws}_3 = 18.3 \text{ m s}^{-1}$). The surface static stability is strong ($N_{\text{sfc}}^2 = 4.4 \times 10^{-4} \text{ s}^{-2}$, quasi-identical to C3), but unlike C3, lapse rates decrease progressively and consistently with altitude ($\Gamma_{400-600} = -0.29 \pm 0.72$, $\Gamma_{600-800} = -0.48 \pm 0.57 \text{ K (100 m)}^{-1}$), indicating a well-stratified column that cools steadily rather than forming a near-isothermal decoupled layer. The surface wind spread ($\sigma = \pm 8.8 \text{ m s}^{-1}$) indicates heterogeneous surface coupling within this cluster. This is the smallest cluster ($n = 39$) and its diagnostics should be interpreted with caution.

8 Wind-Only Partition and ARI

Motivated by the loading structure (Section 7.1), a second k -means model is trained using only u and v , normalised identically to the full-feature model (same z-score scaling, same seed and k). The Adjusted Rand Index (ARI) [8] measures the similarity between two clusterings, where 1 indicates perfect agreement, 0 indicates performance no better than random chance, and negative values indicate worse-than-random agreement.

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \frac{[\sum_i \binom{a_i}{2}][\sum_j \binom{b_j}{2}]}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \frac{[\sum_i \binom{a_i}{2}][\sum_j \binom{b_j}{2}]}{\binom{n}{2}}} \quad (18)$$

The ARI between the two partitions equals 0.718: The two partitions show strong agreement ($\text{ARI} = 0.718$), with an alignment accuracy of 85.5%, meaning approximately 15% of soundings are assigned to different clusters when thermodynamic variables are removed. This confirms that wind structure is the primary organising principle of the six-cluster partition, while thermodynamic variables contribute meaningfully to a non-negligible subset of sounding assignments. This result is consistent with the physical expectation at DDU: wind regime largely determines air

mass origin and therefore thermodynamic properties, but the relationship is not deterministic, and some soundings are only unambiguously classified when both kinematic and thermodynamic information are used. Figure 5 shows both partitions.

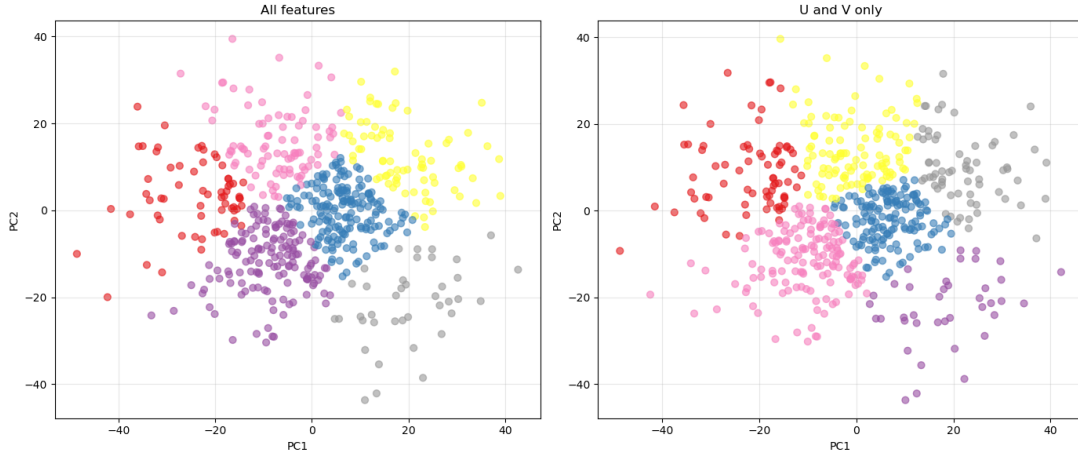


Figure 5: Scatterplot of PCA1 vs PCA2 colored by cluster, seed=24, k=6, for X built with all features vs u and v only

9 Temporal and Inter-Annual Analysis

Figure 6 shows the six-year carpet plot (2020–2025, JJA). Each coloured bar represents one sounding; persistent blocks of the same colour indicate multi-day regime stability; vertically aligned patterns indicate inter-annual recurrence.

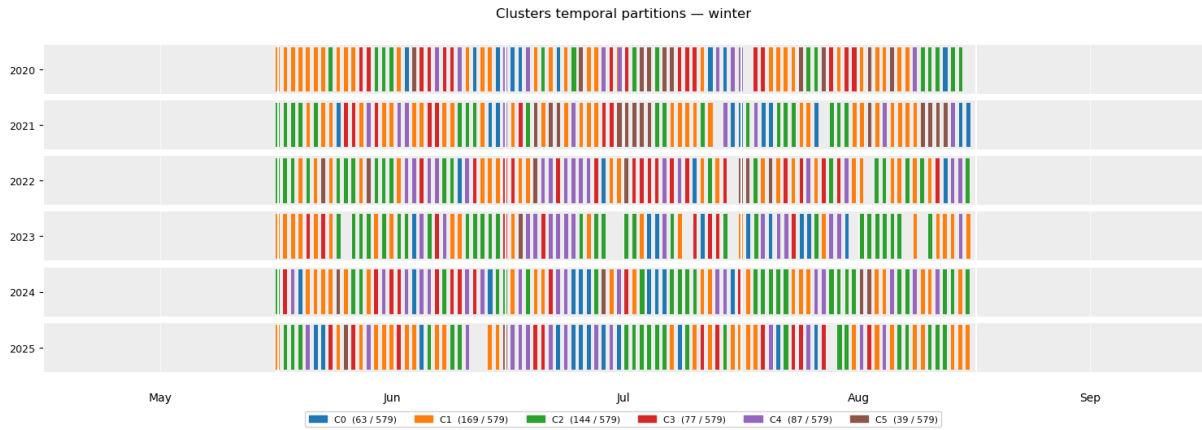


Figure 6: Carpet plot of cluster labels over six austral winters (2020–2025, June–August). Each row is one winter; each bar is one 00:00 UTC sounding coloured by cluster.

Figure 6 shows that most of the time, the clusters are “per group”, suggesting several days of same atmospheric regimes. Under a random partition, cluster labels would be distributed uniformly across the winter months, with no temporal coherence.

Figure 7 shows the inter-annual consistency analysis using the normalised Shannon entropy

$$H_b = -\frac{1}{\log K} \sum_{k=1}^K p_{k,b} \log p_{k,b} \quad (19)$$

per seven-day bin b , where $p_{k,b}$ is the fraction of years in which cluster k is dominant and $K = 6$. $H_b = 0$ means perfect inter-annual consistency; $H_b = 1$ means no consistency (i.e. uniform distribution between all clusters).

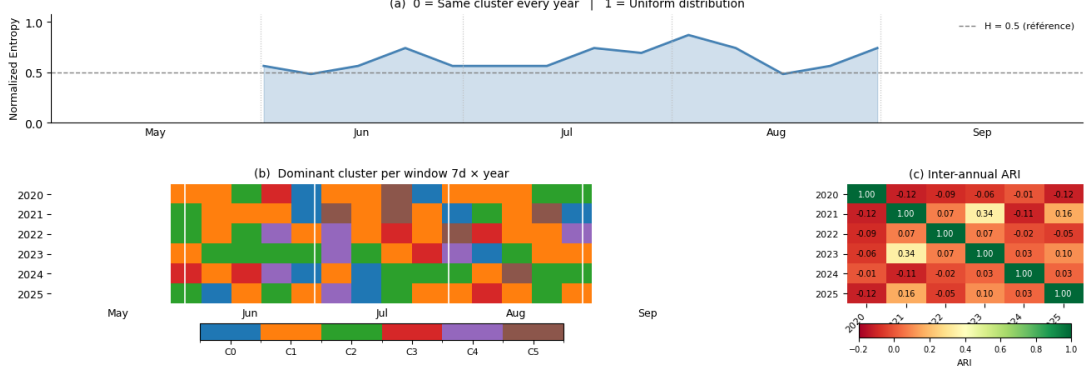


Figure 7: Inter-annual consistency (seven-day bins, 2020–2025). **(a)** Normalised Shannon entropy H_b ; dashed line: $H = 0.5$. **(b)** Dominant cluster per bin (7 days) and year. **(c)** Pairwise ARI between winters.

The entropy varies between 0.5 and 1 showing a partial consistency. In reality when $H = 0.6$ means that ~ 3 clusters share the temporal window across years — visible in Figure 7 b). Therefore we can say that the distribution is not fully random. The inter-annual ARI computes how much the partitions are similar between winters.

10 Discussion

10.1 Physical coherence of the six-regime taxonomy

The six clusters map onto three broad physical forcing categories. It should be emphasised that all physical labels are interpretations derived from *a posteriori* analysis of cluster-mean diagnostics, not causal identifications: the algorithm operates purely on normalised profile distances, and the alignment with physically recognisable patterns is a positive, non-trivial result.

High-wind family (C0, C2). Both clusters exhibit strong surface winds with negative u_{sfc} and positive v_{sfc} , consistent with the 90° – 180° katabatic window documented at DDU [3]. They differ substantially in moisture content: C0 ($\text{IWV} = 5.05 \text{ kg m}^{-2}$, $\overline{RH}_{0-1} = 72\%$) is the most humid cluster, while C2 ($\text{IWV} = 2.68 \text{ kg m}^{-2}$) is moderately humid.

Their thermal structures diverge in a physically instructive way. C0 has a near-zero surface inversion ($\Gamma_{0-200} = +0.18 \text{ K (100 m)}^{-1}$): this is the expected outcome under strong katabatic drainage, where intense mechanical mixing destroys the pre-existing radiative inversion and produces a vertically well-mixed surface layer. C2, by contrast, maintains a well-defined inversion below the jet nose ($\Gamma_{0-200} = +0.65 \text{ K (100 m)}^{-1}$) despite comparable surface wind speeds

($ws_{\text{sfc}} = 14.7 \text{ m s}^{-1}$ vs. 15.3 m s^{-1} in C0). This inversion is not radiative in origin: it is a kinematic structure maintained by the drainage flow itself, which carries cold continental air at low levels beneath warmer free-tropospheric air, creating a temperature increase from the surface to the jet nose. The strongest negative lapse rates of all clusters immediately above the jet nose ($\Gamma_{400-600} = -0.71 \text{ K (100 m)}^{-1}$, with the lowest standard deviation in the entire lapse-rate table) confirm the sharpness and consistency of this two-layer structure across individual soundings. What drives the large moisture difference between C0 and C2 despite their similar surface-wind signatures cannot be determined from profile data alone; their separation in clustering space is primarily kinematic, while the humidity contrast is a diagnostically useful but *a posteriori* property. The z-score table (Table 5) makes this separation explicit: C0 and C2 have opposite signatures on v_3 ($z = -1.40$ vs $+0.12$) and on the lapse-rate windows 400–800 m ($z \approx -0.3$ for C0 vs -0.44 to -0.51 for C2), confirming that their kinematic separation in PCA space reflects both a difference in upper-level meridional flow and in the sharpness of the thermal structure above the jet nose. This is further supported by the z-score on ws_{500} : C0 reaches $z = +1.24$ while ws_{sfc} remains at $z = +0.78$, confirming quantitatively that the wind maximum in C0 lies above the surface rather than at it, the diffuse elevated jet structure that distinguishes C0 from C2.

Strong upper-level flow family (C3, C5). Both clusters are characterised by weak surface winds and a strong jet aloft, and both have dry, stably stratified columns. C3 has the strongest upper-level flow ($ws_3 = 20.3 \text{ m s}^{-1}$, $u_3 = +19.1 \text{ m s}^{-1}$) and the highest surface static stability ($N_{\text{sfc}}^2 = 4.5 \times 10^{-4} \text{ s}^{-2}$), with a near-isothermal layer above 400 m forming a dynamical barrier between the surface and the overlying flow. C5 has the driest column ($\text{IWV} = 1.03 \text{ kg m}^{-2}$, dew-point depression 18.5 K in the 1–3 km layer) and a dominant northward upper-level flow ($v_3 = +16.8 \text{ m s}^{-1}$), consistent with a continental outflow trajectory. The two clusters differ primarily in the direction of the upper-level flow. A notable tension arises in C3: the “decoupled” label reflects the near-isothermal near-surface layer and strong static stability, yet C3 co-occurs with one of the strongest upper-level jets in the dataset.

Quantitatively, C3 is the only cluster with positive z -scores on the lapse-rate windows 400–1000 m ($z = +0.76$ to $+0.97$, Table 5), meaning its 400–1000 m layer is on average warmer relative to the surface than the dataset mean, the statistical signature of the quasi-isothermal decoupled layer. C5, by contrast, has the most extreme z -score in the entire table on v_3 ($z = +1.87$) and the lowest on $\overline{RH}_{1-3}$ ($z = -1.13$), making it the most unambiguously defined cluster in the dataset despite its small size.

Noteworthy, C5 has the most internally homogeneous humidity of all clusters ($\sigma_{\text{IWV}} = 0.31 \text{ kg m}^{-2}$, the lowest in the dataset), which supports the physical reality of this regime despite its small sample size ($n = 39$).

Moderate-to-weak surface wind family (C1, C4). C4 ($\text{IWV} = 4.06 \text{ kg m}^{-2}$, $\overline{RH}_{0-1} = 67\%$) is the second most humid cluster with moderate winds at all levels; its wind turning of $-64^\circ \pm 129^\circ$ reveals substantial directional heterogeneity, suggesting this cluster aggregates physically distinct situations with similar speed profiles but different directional structures. C1 ($\text{IWV} = 1.74 \text{ kg m}^{-2}$) groups weak-wind, moderately dry soundings with equally heterogeneous

turning ($+71^\circ \pm 97^\circ$). For both clusters, directional interpretations should be treated with caution; their thermodynamic and speed-based descriptions remain meaningful. The z-score table further reveals that C4 and C0 share a similar positive signal on IWV ($z = +0.76$ and $+1.31$ respectively) and $\overline{RH}_{0-1}$ ($z = +0.79$ and $+1.06$), but are separated by their upper-level meridional flow: $v_3 = -1.06$ for C4 vs -1.40 for C0 at surface level, and opposite signs at 3 km. Their humidity similarity and directional divergence explain why C4 is difficult to characterise beyond its moisture content.

The distribution of cluster sizes reflects both the physical variability of DDU winters and the resolution limits of the six-cluster solution. C1 is the largest cluster ($n = 169$, 29%), but its absence of any discriminating signal ($|z| < 0.5$ on all variables, Table 5) and its large directional heterogeneity ($\pm 97^\circ$) indicate that it acts as a statistical residual aggregating weakly-forced situations that $k = 6$ cannot resolve, rather than representing a single coherent physical regime. Its large size should therefore be interpreted as a limitation of the partition rather than evidence that calm dry conditions dominate DDU winters. C2 is the second largest ($n = 144$, 25%), consistent with the well-documented frequency of moderate katabatic drainage at this site. C5 is the smallest cluster ($n = 39$, 7% of the retained soundings), suggesting that strong dry northward upper-level flow may be an episodic feature of DDU winters, though no independent climatological estimate is available in the literature to confirm this.

10.2 On the choice of k

The global silhouette optimum at $k = 2$ ($\bar{s} = 0.30$) correctly identifies the active/passive wind dichotomy. However, a single active cluster would conflate C0, C2, and C4, three states with different moisture contents and tropospheric-layer structures, providing insufficient physical resolution. Among the local secondary optima identified over 500 seeds, $k = 6$ is the most frequent (176/500), ahead of $k = 8$ (163/500) and $k = 5$ (158/500), and is therefore retained.

This choice remains somewhat arbitrary: $k = 8$ is statistically nearly as well supported and would potentially separate sub-regimes currently merged in C1 and C4, both of which exhibit large directional heterogeneity. The decision to stop at $k = 6$ reflects a pragmatic trade-off between physical resolution and statistical robustness: at $k = 6$ the smallest clusters already have limited sample sizes (C5: $n = 39$; C3: $n = 77$), and increasing k globally would degrade the reliability of cluster-mean diagnostics. The two-stage thermodynamic refinement (Appendix 13) provides a principled alternative by applying additional splitting only where internal thermodynamic structure warrants it.

10.3 Dependence on normalisation and interpretive limitations

The wind-only ARI of 0.718 confirms that wind structure is the dominant organising principle, but this result is self-consistent rather than independent: the normalisation already encodes the assumption that wind dominates by giving wind components larger effective variance in PCA space ($\sigma_u = 1.22$, $\sigma_v = 1.03$ versus $\sigma \approx 0.12$ – 0.17 for thermodynamic variables). A balanced normalisation would likely produce a different partition, potentially more thermodynamically defined. The ARI should therefore be read as a validation of internal consistency, not as an independent test of the physical hypothesis.

Despite these limitations, the overall coherence of the partition is good. The kinematic inversion structure in C2, the near-isothermal decoupled layer in C3, the extreme humidity of C0, and the extreme dryness and internal homogeneity of C5 all correspond to physically distinct and recognisable atmospheric states. The bootstrap stability (96.2% of pairs stably assigned) and the temporal clustering of regime occurrences (Section 9) further support the robustness of the partition, suggesting the six clusters capture recurring atmospheric structures at DDU rather than artefacts of the method.

11 Limitations and Perspectives

11.1 Limitations

Selection bias from the coverage filter. The coverage filter discards 21.8% of raw soundings, retaining only profiles that start below 50 m and reach at least 85% of the grid top (≈ 4300 m). Rejected profiles correspond disproportionately to extreme conditions (balloon failure during blizzards), so the frequencies of the most intense katabatic clusters (C0, C2) are likely underestimated. The reported proportions characterise the population of *well-sampled* soundings, not all winter atmospheric states.

Restriction to 00:00 UTC launches. The analysis retains only the 00:00 UTC soundings, corresponding to 10:00 local time at DDU. The 12:00 UTC launches (22:00 local time) are excluded. Extending the dataset to include these evening soundings would allow the diurnal cycle of each regime to be characterised, though it would require an appropriate correction to avoid artefacts from mixing two phases of the katabatic cycle.

Directional heterogeneity in weakly-constrained clusters. The within-cluster standard deviation of wind turning exceeds 90° in C1 ($\pm 97^\circ$) and C4 ($\pm 129^\circ$). This is not a computational artefact: the turning is computed as a proper circular difference and is well defined for each sounding individually. It reflects the fact that the k -means distance metric groups situations that differ substantially in their directional structure. Directional interpretations for C1 and C4 should therefore be treated with caution.

Inferential physical attribution. The interpretation of each cluster rests entirely on a *posteriori* analysis of cluster-mean diagnostics without external validation against synoptic reanalyses. Some clusters may aggregate meteorologically distinct situations that produce similar profile shapes. ERA5 synoptic composites (sea-level pressure, 500-hPa geopotential, meridional moisture flux) would provide a largely independent validation.

Normalisation and kinematic dominance. The hybrid normalisation intentionally gives wind components larger variance in PCA space than thermodynamic variables ($\sigma_u = 1.22$, $\sigma_v = 1.03$ vs $\sigma \approx 0.12$ – 0.17 for thermodynamics). This encodes the assumption that wind structure is the primary discriminant, which is confirmed by the wind-only ARI of 0.718: the partition is predominantly kinematic, but thermodynamic variables contribute to approximately 15% of

sounding attributions. Any analysis that relies on the thermodynamic characteristics of the clusters should bear in mind that these variables had a secondary role in defining them.

Limited sample size and record length. After filtering, 579 soundings over six winters are available. The smallest clusters (C5: $n = 39$; C3: $n = 77$) have limited statistical robustness, and six years are insufficient to separate interannual variability from longer-term trends.

11.2 Perspectives

Validation against ERA5 synoptic composites. Compositing ERA5 sea-level pressure, 500-hPa geopotential, and meridional moisture flux by cluster label would confirm the physical attributions, for example verifying that C0 corresponds to a specific large-scale pressure configuration, without the circularity of comparing ERA5 profiles at the DDU grid point directly.

Climatological trends and interannual variability. Extending the dataset to 10–20 years would allow detection of trends in cluster frequency. An increase in humid intrusions (C0, C4) would be consistent with Southern Ocean warming and projected poleward moisture transport increases; changes in katabatic regimes (C0, C2) could be linked to surface mass balance trends.

Link to surface precipitation and sublimation. Crossing cluster labels with surface precipitation and blowing-snow records at DDU would quantify each regime’s contribution to the local surface mass balance. The working hypothesis is that C0 drives the most intense precipitation events while C2 maximises sublimation[7] and snow drift.

Comparison with other East Antarctic coastal stations. Applying the same methodology to Mawson and Casey would test whether the six-regime taxonomy is specific to DDU or generalises across the coastal margin, separating the local katabatic signature from large-scale circumpolar forcing.

Near-real-time classification. With fixed k -means centroids, any new sounding can be classified instantaneously in PCA space, enabling regime-conditioned operational forecasts of surface wind, precipitation, and visibility at DDU.

Inclusion of evening launches. Including the 12:00 UTC launches (22:00 local time) after subtracting a climatological diurnal anomaly at each level would double the dataset to approximately 1150 soundings and allow the evening tropospheric layer to be studied within the same classification framework, providing a more complete picture of the diurnal cycle of each regime.

12 Conclusion

Six years of DDU winter radiosondes have been classified into six atmospheric regimes using k -means on hybrid-normalised, PCA-reduced concatenated profiles (global min-max for thermodynamic variables, block z-score for wind components).

The global silhouette optimum at $k = 2$ ($\bar{s} = 0.30$) correctly identifies the active/passive wind dichotomy but provides insufficient physical resolution. Among the local secondary optima identified over 500 seeds, $k = 6$ is the most frequent (176/500), and is therefore retained as the primary result. Bootstrap stability confirms that 96.2% of sounding pairs are stably assigned. A wind-only partition yields $\text{ARI} = 0.718$, indicating that the six regimes are predominantly kinematically defined, with thermodynamic variables contributing to approximately 15% of sounding attributions.

The six regimes are: humid katabatic–synoptic coupling (C0), weak-wind dry state (C1), katabatic–synoptic coupling (C2), decoupled layer (C3), humid moderate-wind regime (C4), and dry meridional outflow (C5). The katabatic clusters C0 and C2 together account for 36% of soundings ($n = 207$). C1 is the largest cluster ($n = 169$, 29%); its size reflects the aggregation of weakly-forced situations unresolved at $k = 6$ rather than the dominance of a single coherent regime. C5 is the smallest regime ($n = 39$, 7%), suggesting that dry northward upper-level flow may be episodic at DDU, though no independent climatological estimate is available to confirm this. C0 carries the highest column moisture ($\text{IWV} = 5.05 \text{ kg m}^{-2}$) and is the primary candidate for high-impact precipitation events at DDU, though a direct link to surface observations remains to be established.

A two-stage thermodynamic refinement, presented in Appendix 13, sub-divides four of the six wind regimes and yields 11 sub-clusters. The most significant subdivision occurs within the katabatic–synoptic coupling regime (C2, KSC; silhouette $s = 0.465$), separating a dry sub-type ($\text{IWV} = 1.19 \text{ kg m}^{-2}$) from a humid one ($\text{IWV} = 3.23 \text{ kg m}^{-2}$). The dry meridional outflow regime (C5, DMO) also separates into a dry ($\text{IWV} = 1.93 \text{ kg m}^{-2}$) and a humid sub-type ($\text{IWV} = 4.05 \text{ kg m}^{-2}$). The two smallest regimes (C0, HKS, $n = 63$; C1, WWD, $n = 169$) could not be meaningfully subdivided given the sample size constraint for HKS and the directional heterogeneity of WWD.

Data and Code Availability

Radiosonde data are publicly available through the Météo-France open data portal [1]. The clustering pipeline is implemented in the `ClusterProfiles` class (`cluster_profile.py`), with methods `build_X`, `treat_nan`, `normalize_shape` (global min-max for T , T_d , $T - T_d$) followed by `normalize_block` (z-score for u , v), `apply_pca` ($n_components = 4$), `fit_kmeans` (`seed=24`, `n_init='auto'`), `plot_cluster_carpet`, and `plot_interannual_consistency`. Dependencies: `scikit-learn`, `scipy`, `numpy`, `pandas`, `matplotlib`.

Code is available on [Github](#)

References

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13 Appendix

13.1 Two-Stage Thermodynamic Refinement

13.1.1 Procedure

The six-cluster partition is predominantly kinematic ($\text{ARI} = 0.718$ between full-feature and wind-only partitions, Section 8). Within each wind regime, thermodynamic variability that the primary clustering only partially resolves may contain physically meaningful sub-structure. The two-stage approach addresses this systematically.

Stage 1 applies k -means ($k_{\text{wind}} = 6$, $n_{\text{pca,wind}} = 4$, seed 24, $n_{\text{init}} = \text{auto}$) to the wind-only sub-matrix, recovering the six wind regimes. Stage 2 sub-divides each regime w thermodynamically: PCA is applied to the thermodynamic submatrix (T , T_d , $T - T_d$) of regime w ; $n_{\text{pca}}^{(w)}$ is the smallest number of components reaching $\geq 90\%$ cumulative variance; $k_{\text{therm}}^{(w)}$ is selected by silhouette maximisation over $k \in [2, 5]$, subject to silhouette > 0.25 and all sub-clusters ≥ 10 soundings.

13.1.2 Results

table 7 summarises the procedure. Four of the six wind regimes are sub-divided, while regimes 0 and 1 could not be subdivided because all candidate sub-clusters contained fewer than 10 soundings. This yields **11 combined clusters**.

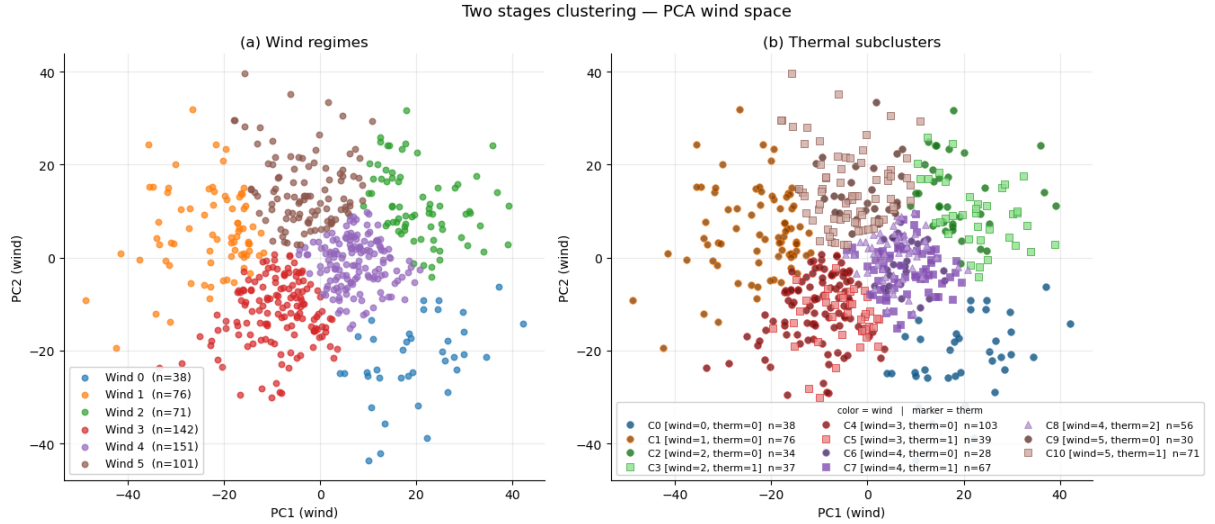
Table 7: Two-stage clustering results (seed 24, $n_{\text{init}} = \text{auto}$). Wind regime labels are approximate matches to the primary six-cluster partition, identified by kinematic signature from the wind-only partition; the match for HMW (wind regime 4) is approximate due to IWV divergence when thermodynamic variables are excluded. Regimes 0 (HKS) and 1 (WWD) are not subdivided; rejection reasons are detailed in the last column.

Wind regime	n_w	Approx. label	n_{pca}	Silhouettes ($k = 2 \dots 5$)	k_{therm}	Rejection reason
0	38	HKS	5	0.484, 0.269, 0.237, 0.258	1	all k : min < 10
1	76	WWD	7	0.440, 0.421, 0.335, 0.215	1	$k=2$: min=9 < 10 ; $k \geq 3$: min < 10
2	71	DMO	7	0.268, 0.226, 0.205, 0.208	2	—
3	142	KSC	5	0.465 , 0.384, 0.289, 0.306	2	—
4	151	HMW	7	0.259, 0.260 , 0.216, 0.192	3	—
5	101	DL	6	0.345 , 0.309, 0.282, 0.266	2	—
Total					11 clusters	

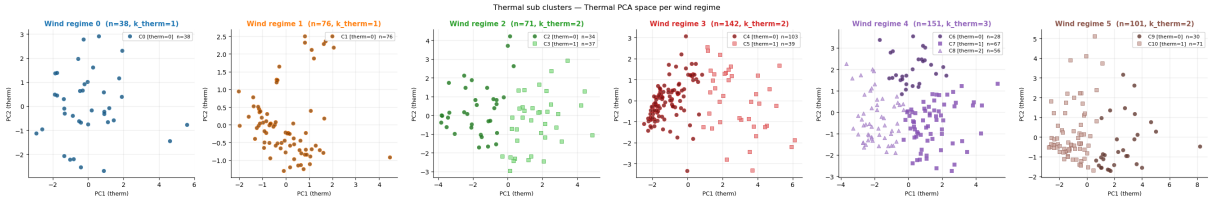
The eleven combined clusters, their parent wind regime, and IWV are: C0 (wind 0/HKS, $n = 38$, $\text{IWV} = 1.04 \text{ kg m}^{-2}$), C1 (wind 1/WWD, $n = 76$, $\text{IWV} = 5.19 \text{ kg m}^{-2}$), C2 (wind 2/DMO-A, $n = 34$, $\text{IWV} = 2.69 \text{ kg m}^{-2}$), C3 (wind 2/DMO-B, $n = 37$, $\text{IWV} = 1.20 \text{ kg m}^{-2}$), C4 (wind 3/KSC-humid, $n = 103$, $\text{IWV} = 3.23 \text{ kg m}^{-2}$), C5 (wind 3/KSC-dry, $n = 39$, $\text{IWV} = 1.19 \text{ kg m}^{-2}$), C6 (wind 4/HMW-A, $n = 28$, $\text{IWV} = 1.64 \text{ kg m}^{-2}$), C7 (wind 4/HMW-B, $n = 67$, $\text{IWV} = 1.15 \text{ kg m}^{-2}$), C8 (wind 4/HMW-C, $n = 56$, $\text{IWV} = 2.26 \text{ kg m}^{-2}$), C9 (wind 5/DL-dry, $n = 30$, $\text{IWV} = 1.93 \text{ kg m}^{-2}$), C10 (wind 5/DL-humid, $n = 71$, $\text{IWV} = 4.05 \text{ kg m}^{-2}$).

The most significant thermodynamic subdivision occurs in wind regime 3 (KSC, $n = 142$), which achieves the highest within-regime silhouette ($s = 0.465$), separating a humid sub-type (C4, $n = 103$, $\text{IWV} = 3.23 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 57\%$) from a dry sub-type (C5, $n = 39$, IWV

$= 1.19 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 33\%$). Wind regime 5 (DBL, $n = 101$) separates a humid sub-type (C10, $n = 71$, $\text{IWV} = 4.05 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 59\%$) from a dry sub-type (C9, $n = 30$, $\text{IWV} = 1.93 \text{ kg m}^{-2}$, $\overline{\text{RH}}_{0-1} = 60\%$); the near-identical RH values suggest the separation is driven primarily by the 1–3 km layer, where the two sub-types diverge substantially ($\overline{\text{RH}}_{1-3} = 70\%$ vs 52%). Wind regime 4 (HMW, $n = 151$) requires three thermodynamic sub-clusters, with IWV of 1.64, 1.15, and 2.26 kg m^{-2} , consistent with the directionally heterogeneous character of this regime already noted in the primary partition. Wind regime 2 (DMO, $n = 71$) splits into two sub-types of comparable size ($n = 34$ and $n = 37$) with IWV of 2.69 and 1.20 kg m^{-2} respectively, a humidity contrast of similar magnitude to that observed in the KSC regime.

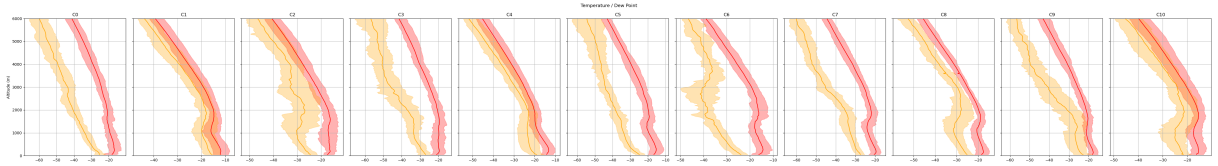


(a) Clustering using PCA on wind features u and v only, with markers showing the sub-clusters obtained from thermodynamic features — representation in the PCA_{wind} space.

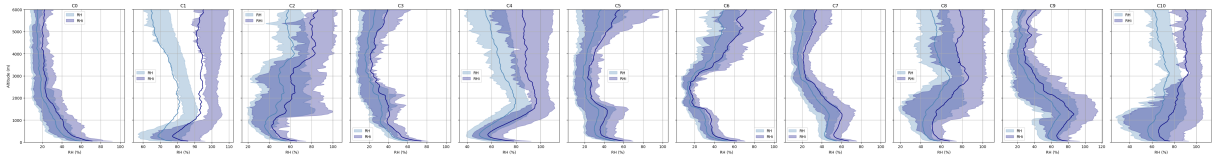


(b) Clustering of wind clusters with respect to PCA on thermodynamic variables T , $T - T_d$, and T_d — representation in the $\text{PCA}_{\text{therm}}$ space.

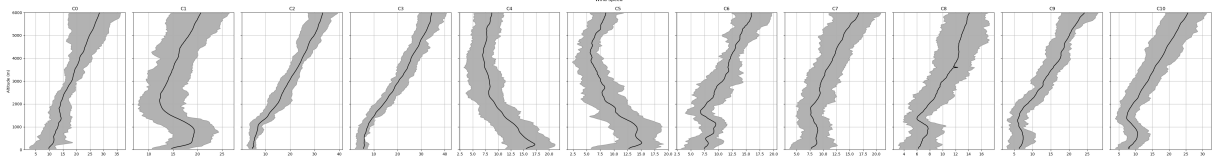
Figure 8: Two-stage clustering: PCA coloured per cluster with respect to PCA on thermodynamic variables (b) and wind variables (a).



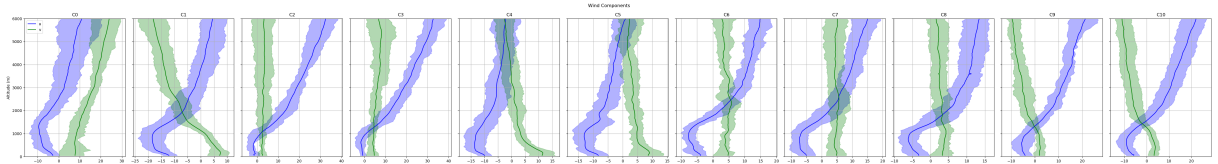
(a) Temperature (T , red) and dew-point temperature (T_d , orange) profiles for the eleven clusters.



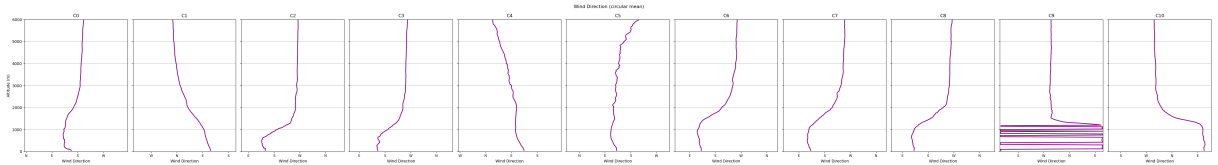
(b) Relative humidity (RH and RH_i) profiles for the eleven clusters.



(c) Wind-speed (WS) profiles for the eleven clusters.



(d) Zonal (u , solid) and meridional (v , dashed) wind-component profiles for the eleven clusters.



(e) Wind-direction (dd) profiles for the eleven clusters.

Figure 9: Cluster-mean vertical profiles (50–6000 m) for the eleven clusters. Panels show temperature and dew point (T , T_d), relative humidity (RH), wind speed (WS), zonal and meridional wind components (u , v), and wind direction (dd). Shaded ribbons indicate the 25th–75th percentile range.